The Electronic Grip Gauge (EGG): Automated Assessment of Sensorimotor Hand Function Using an Instrumented Fragile Object

Michael D. Adkins, Tyler J. Gourley, Tyler S. Davis, Nate Toth, Monika K. Buczak, *Graduate Student Member, IEEE*, Masaru Teramoto, Steven R. Edgley, Lorie G. Richards, Marta M. Iversen, and Jacob A. George, *Member, IEEE*

***Abstract*— Hand dexterity assessments play a crucial role in informing the rehabilitative care of individuals with upper-limb hemiparesis. However, current assessments often struggle to evaluate the hand's ability to precisely control grip force, a skill vital for daily activities like handling fragile objects. Here we describe the design of the Electronic Grip Gauge (EGG), an adjustable-weight, instrumented "fragile" object that measures grip force, load force, acceleration, orientation, and relative position. Embedded sensors enable automatic segmentation and analysis of EGG transfers in various modes. In "Non-Fragile" mode, there is no break threshold; the EGG serves as an automated variant of the Box-and-Blocks test. In "Fragile" mode, the EGG simulates fragility by playing a "break" noise if grip force exceeds a set threshold, requiring grip control to prevent breaks. In "Fragile-Feedback" mode, audio-visual feedback is provided proportional to applied grip force to supplement potentially impaired tactile feedback. Demonstrating functionality, we evaluated sensorimotor differences between 26 hemiparetic and 26 age-matched healthy participants. In "Fragile" mode, paretic hands were significantly slower, applied excessive force, and broke the EGG more frequently than contralateral and healthy control hands. In "Fragile-Feedback" mode, a subset of paretic hands improved, transferring the EGG faster and/or with less force. This work demonstrates the EGG's utility in automatically quantifying sensorimotor deficits and that, for a subset of hemiparetic patients, audiovisual feedback could potentially coach and rehabilitate hand function. Collectively, this work showcases the EGG's potential as both an assessment and rehabilitation device for grip force control–a critical skill in hand therapy.**

***Index Terms*— Stroke, hemiparesis, sensorimotor function, grip force regulation, hand dexterity, tactile feedback, fragile object.**

Received 17 October 2025; revised 17 February 2026; accepted 17 March 2026. Date of publication 10 April 2026; date of current version 1 May 2026. Associate Editor: Claudio Castellini. This work was supported in part by the Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD) through the National Institutes of Health (NIH) under Award 1F30HD117488-01A1, in part by the Office of the Director (OD), NICHD, and National Institute of Dental and Craniofacial Research (NIDCR) through NIH under Grant DP5OD029571 and Grant DP5OD029571S01, and in part by the University of Utah's Department of Physical Medicine and Rehabilitation through an internal pilot at the Craig. H. Neilsen. *(Corresponding authors: Michael D. Adkins; Jacob A. George.)*

Michael D. Adkins and Tyler J. Gourley are with the Department of Electrical and Computer Engineering, University of Utah, Salt Lake City, UT 84112 USA (e-mail: michael.adkins@hsc.utah.edu).

Tyler S. Davis is with the Department of Neurosurgery, University of Utah, Salt Lake City, UT 84112 USA.

Nate Toth is with the Department of Mechanical Engineering, University of Utah, Salt Lake City, UT 84112 USA.

Monika K. Buczak is with the Department of Biomedical Engineering, University of Utah, Salt Lake City, UT 84112 USA.

Masaru Teramoto, Steven R. Edgley, and Marta M. Iversen are with the Department of Physical Medicine and Rehabilitation, University of Utah, Salt Lake City, UT 84112 USA.

Lorie G. Richards is with the Department of Occupational and Recreational Therapies, University of Utah, Salt Lake City, UT 84112 USA.

Jacob A. George is with the Department of Electrical and Computer Engineering, the Department of Physical Medicine and Rehabilitation, the Department of Biomedical Engineering, and the Department of Mechanical Engineering, University of Utah, Salt Lake City, UT 84112 USA (e-mail: jacob.george@utah.edu).

This article has supplementary downloadable material available at https://doi.org/10.1109/TNSRE.2026.3683004, provided by the authors.

Digital Object Identifier 10.1109/TNSRE.2026.3683004

I. Introduction

UPPER-LIMB hemiparesis is a common and profoundly debilitating impairment that is frequently acquired as a result of a neurological injury such as a stroke, traumatic brain injury (TBI), or cerebral palsy [1], [2], [3]. Upper-limb hemiparesis is characterized by weakened control of the hand and arm on one side of the body, and may be further accompanied by pain, deficits in proprioception, and/or loss of ability to perceive touch (tactile feedback) in the affected limb [4], [5]. These sensorimotor impairments can deeply impact an individual's quality of life, reducing their ability to complete everyday activities of daily living such as eating, dressing, and self-care [6], [7]. Upper-limb hemiparesis is especially prevalent among stroke survivors, with roughly 80% experiencing impairment acutely, and 40% experiencing it chronically [1].

Although spontaneous recovery of hand function is possible following a neurological injury, it is well-established that guided, patient-specific neurorehabilitation significantly improves the chances of regaining lost function [1], [8]. Accurate assessments of hand function, therefore, play a critical role in improving patient rehabilitation, as they not only allow for rehabilitation to be better personalized to a patient's unique sensorimotor profile, but also allow for the efficacy of an ongoing intervention to be evaluated [9], [10].

However, standard-of-care assessments frequently neglect and overlook a key aspect of hand function: the ability to precisely control grip force [11], [12]. Established upper-limb motor tests such as the Box and Blocks Test (BBT) [13], Nine-Hole Peg Test (9HPT) [14], and Purdue Pegboard Test (PPT) [15] evaluate motor function in terms of the ability to transfer objects quickly from one location to another. Although these tests provide useful metrics for gross motor function and manual dexterity, they do not directly measure the finer sensorimotor aspects of hand function, such as the ability to finely regulate grip force. Similarly, existing sensory assessments, such as the Two-Point Discrimination test (2PDT) [16] for spatial discrimination, and the Semmes-Weinstein monofilament test (SWMT) [17] for tactile perception, evaluate sensory function in isolation, ignoring the sensorimotor integration necessary for completing fine sensorimotor tasks.

In an effort to better assess the fine sensorimotor function of the hand, researchers have created variants of the BBT that utilize fragile objects with known "break" thresholds. Here, users are asked to complete a fragile object transfer or lifting task, where they must grasp a fragile object with one hand and avoid breaking the object as they transfer it over a short barrier [18] or lift it vertically to a set height [19]. This task requires precise grip force regulation as the user's applied grip force must exceed the minimum necessary to pick up the object while also staying below the "break" threshold [20]. Accompanying these fragile object tasks, several fragile objects have been used, such as paper blocks held together lightly with toothpicks [21], metal plates separated by a weak magnetic field [22], [23], [24], and collapsible 3D-printed blocks embedded with magnets [25], [26], [27]. In newer adaptations, studies have utilized instrumented handles, cylinders, and 3D-printed blocks, which integrate load cells and accelerometers to directly measure applied grip force and object accelerations [18], [20], [28], [29], [30], [31], [32]. Importantly, utilizing fragile objects to both assess and potentially rehabilitate hemiparetic hand function remains an area of ongoing and active research [19], [32], [33].

Building on these prior works, we introduce the Electronic Grip Gauge (EGG), an instrumented 3D-printed block that can simulate a fragile object. The EGG is embedded with sensors that allow it to measure applied grip force, vertical load force, acceleration, angular velocity, magnetic fields, device orientation, and relative object position in real time. The EGG is a non-collapsible rigid object that mimics fragility by playing a break sound when the applied grip force exceeds a tunable break threshold. The EGG shares its closest similarities with prior instrumented devices, which evaluate fine grip force control and mimic fragility [29], [30], [32]. Distinct from prior work, the EGG utilizes a novel position tracking system to fully automate the analysis of transfers. As a user moves the EGG over a short barrier, it automatically identifies the transfer and calculates key performance metrics (e.g., transfer duration, peak grip force, if the break force threshold was exceeded, etc.). This performance data is then immediately displayed to the end users' connected device (phone, computer, tablet, etc.) via a custom app, which the EGG connects to wirelessly over Bluetooth. Further, the EGG introduces a novel audiovisual feedback system for supplementing tactile feedback and assessing sensory versus motor deficits.

In this manuscript, we first outline the mechanical and electrical design of the EGG and then describe the software approach towards automated transfer labeling and performance metric generation. We also detail the EGG's three distinct operating modes: 1) Non-Fragile, 2) Fragile, and 3) Fragile-Feedback (a fragile mode, which provides audiovisual feedback proportional to the users' applied grip force). These three modes are designed to test different aspects of sensorimotor hand function, helping to potentially disentangle sensory deficits from motor deficits. Improved performance with supplemental audiovisual feedback could indicate that the underlying impairment is predominantly sensory in nature. Finally, we demonstrate the usability of the EGG by evaluating the sensorimotor hand function of 26 patients with upper-limb hemiparesis, relative to 26 age-matched healthy controls. In addition to EGG performance metrics, supplementary results from screening assessments for motor function, tactile sensory function, strength, spasticity, and proprioception provide general insight into the sensorimotor profile of the recruited hemiparetic group.

The EGG is available online at www.egg.rehab. It is our hope that by creating a device that can provide high-resolution automatic performance metrics for fine sensorimotor hand function, we can enable better, more personalized hand rehabilitation for patients with upper-limb hemiparesis, as well as better assess the efficacy of ongoing interventions.

II. Device Development

A. Motivation and Design Criteria

The EGG was developed to address the need for a more convenient, reliable, and standardized tool for assessing the hands' fine sensorimotor function and ability to control grip force. Prior approaches for assessing grip force control have generally fallen into two categories: analog fragile objects (e.g., real-life fragile objects [34], and mechanical collapsible objects [23], [24], [35]), and instrumented laboratory devices that are typically rigid (e.g., sensorized handles and instrumented cubes [30], [32]). These approaches have come with unique limitations.

Real-life fragile objects, while ecologically valid, break permanently and lack standardization due to natural variation in size, shape, and break thresholds, making it difficult to compare performance between patients and track individual improvements over time. Analog/mechanical devices, while useful, often only provide binary (break/no break) data, may lack adjustable difficulty, and do not provide continuous data on how applied grip forces evolve over time during object handling (limiting the ability to ascertain metrics such as peak grip force and average grip force used during a transfer). Critically, both real and analog fragile objects require extensive manual data collection, usually by a secondary observer with a stopwatch who notes transfer durations and breaks, limiting clinical efficiency and individual unsupervised home use [23], [24], [34].

Conversely, rigid, instrumented objects, while able to provide continuous data on how grip force evolves over the course of a transfer, are often designed for strictly controlled laboratory settings, lacking portability and ease of use. They may be tethered to external data-acquisition systems or power sources, which can restrict the range of motion during object transport tasks, and they generally require offline data processing, preventing immediate user feedback [19], [36]. These aspects limit clinical translation and home usage.

To address many of the identified shortcomings from both analog and instrumented approaches, and provide a greater level of functionality, the EGG was conceived as a novel, wireless instrumented device, meeting several important design criteria: 1) A standardized shape and size to ensure consistency in performance metric comparison between patients and across time points. 2) High-speed wireless data streaming to provide a convenient cable-free user experience while capturing continuous data on applied grip force. 3) Automatic data analysis, to improve ease of use in the clinic and enable unsupervised home usage (no need for a secondary observer to handle manual data recording). 4) High-resolution quantitative performance metrics, which are immediately displayed to the user (via a real-time app). 5) Adjustable difficulty, in the form of a fully programmable break threshold and swappable weight tray, enabling secondary potential as a rehabilitation device (repeated practice on a task which appropriately challenges patients is a key aspect of stimulating neurorehabilitation [37]). 6) Potential for sensorimotor training using audiovisual feedback to coach patients as they practice handling a fragile object. In the following sections, we elaborate in greater detail on how each of these criteria is satisfied.

B. Device Design: Electronic Grip Gauge

At the heart of the EGG's design is a fully custom, compact printed circuit board (PCB) (18 mm ×34 mm). This PCB integrates the following key components and features: three instrumentation amplifiers (ADS1220, Texas Instruments) for load cell force sensing, a nine-degree-of-freedom inertial measurement unit (9-DOF IMU) (ICM-20948, TDK, Invensense) for orientation and 3-axis sensing of acceleration, angular velocity and magnetic fields, a battery level monitor (MAX 17048, Analog Devices), and a microcontroller with integrated antenna (ESP32-S3-MINI-1, Espressif) alongside other general circuitry (e.g., USB-C connector, battery charging, Light Emitting Diode (LED) logic controller, 3.3 V voltage regulator, power switching, etc.). This PCB then plugs into three 100-N load cells (GML670, Galoce), a Hall-effect sensor (SS49E, Honeywell), an LED feedback light bar (RGWB Neo Pixel Stick, Adafruit), and a 600 mAh 3.7 V rechargeable lithium polymer battery. All these components are then housed within a 3D-printed polylactic (PLA) shell (Fig. 1). Notably, while the EGG's frame is limited to three load cells, our PCB can connect up to 6 separate load cells,

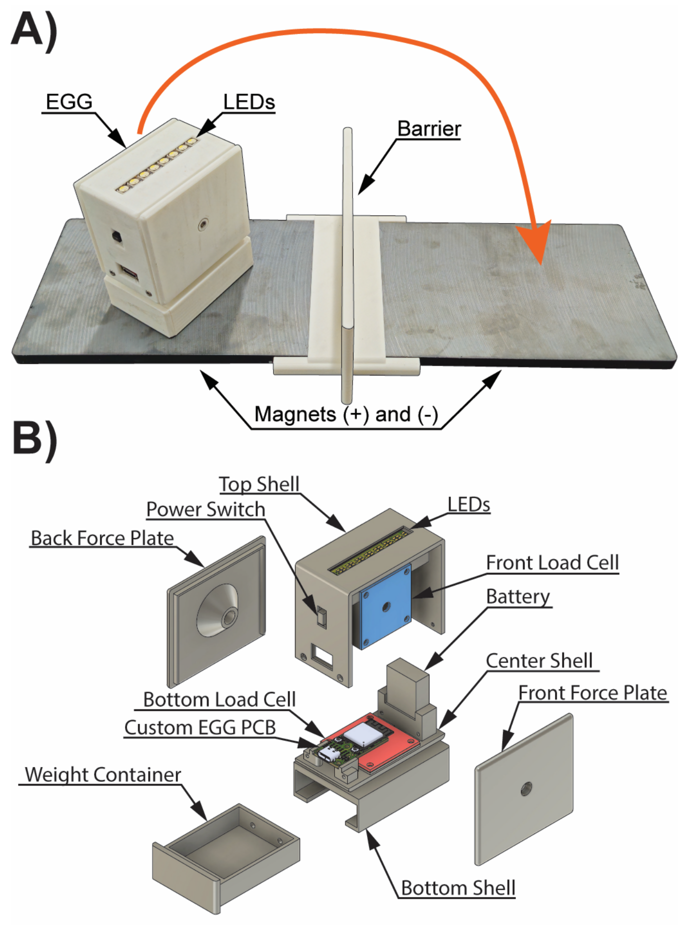

Fig. 1. The Electronic Grip Gauge (EGG). (A) Physical setup: EGG, 2 magnets of opposite polarity, and a 2 inch tall dividing barrier. Participants are tasked with transferring the EGG over the barrier from one magnet to another. Load cells in the front and back of the EGG capture applied grip force during transfers. A bottom load cell captures vertical load (lifting) force. A Hall Effect sensor within the EGG measures magnetic field strength allowing us to track the relative vertical position of the EGG (via the magnitude of the magnetic field strength) and the relative horizontal position of the EGG (via the polarity of the magnetic field) in relation to the magnets. (B) Exploded computer model of the EGG device and its components.

enabling future designs and applications that may require more force sensors.

The assembled EGG measures roughly 66×44.5×81.5 mm. The base of the EGG also incorporates a sliding weight container which houses swappable lead weights, allowing the user to vary the total device weight between 166 and 320 grams. The EGG can be charged while running and reprogrammed via the USB-C port located on the side next to an on/off switch.

C. Data Acquisition, Resolution, and Battery Life

As users interact with the EGG, data from the three load cells (front force, back force, vertical load force), the 9-DOF IMU, and a Hall-effect sensor (magnetic field strength) are wirelessly transmitted over Bluetooth to a nearby computing device (e.g., laptop, tablet, cellphone, etc.) at over 900 Hz (Fig. 2A). This transmitted data is then displayed and analyzed via a custom EGG app loaded on the computing device.

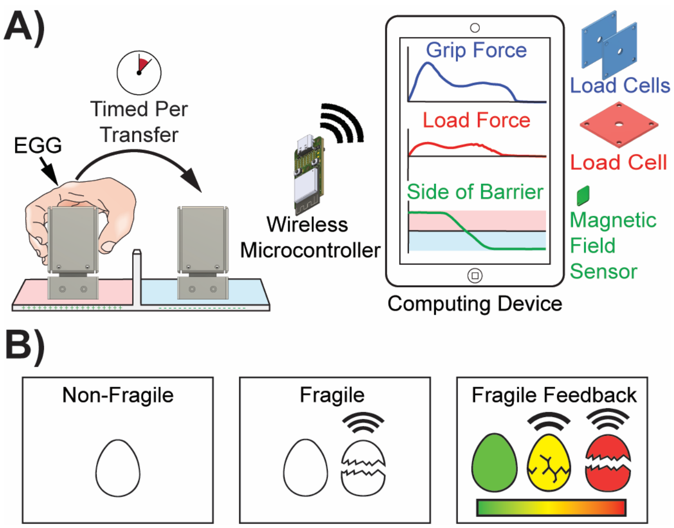

Fig. 2. (A) EGG Transfer Task: As participants transfer the EGG over a barrier, data is transmitted wirelessly to a secondary computing device such as a phone, tablet, or computer for analysis. (B) EGG Modes: During Non-Fragile Mode, any grip force may be used to complete transfers. During Fragile Mode, a break sound is generated if the applied grip force on the EGG exceeds a set break threshold, and the user must regulate their applied grip force. During Fragile-Feedback mode, fragile transfers are given assistive feedback in the form of visual (LED) and sound feedback that is proportional to the applied grip force and indicates how close the EGG is to breakage.

In our testing, we utilized the EGG and a cell phone (Galaxy S23+, Samsung) running this custom app. The custom EGG app was built using Google's open-source Flutter framework and is cross-compilable to all major platforms (iOS, Android, Windows, Linux, macOS, web browsers) with minimal code differences. Grip force applied to the EGG is calculated by taking the average of the front and back force (1).

$$Grip\ Force = (Front\ Force + Back\ Force)/2 \tag{1}$$

The resolution of the applied grip force data depends on factors such as the instrumentation amplifiers' configured sampling rate and gain, the load cell specifications, and the analog circuit design. To maximize accuracy, we chose 100-N load cells with a full-scale error of $\pm$ 0.05% (i.e., $\pm$0.05N error), and in our PCB circuit, we implemented differential low-pass resistor capacitor filters, a power supply bypass capacitor, and referenced sensors to supply voltage. Each ADS1220 instrumentation amplifier was configured to provide its maximum sampling rate of 2,000 samples per second (SPS) and maximum gain of 128. Force sensors were tared and then calibrated using a 200 g weight to output data in Newtons. Under these conditions, we measured sensor noise with a 200g weight placed centrally on the EGG's front force plate and, over the course of 3 minutes, collected 166,221 readings (giving a Bluetooth sampling rate of 923 Hz, with packet loss of 1.7%). From these 166,221 datapoints, we then calculated the errors (e.g., differences between the measured and expected value), and quantified the max error and the standard deviation of errors. The max error is the worst-case maximum amount of error possible in any single force reading, while the standard deviation error represents the typical amount of error present (68% of readings will have this amount of error or less). The front force sensors' max error was $\pm$ 0.16 N, and the standard deviation error was $\pm$ 0.04 N. The other force sensors were also tested and had errors that were the same or lower.

If greater accuracy is required while using the EGG, the ADS1220's returned sampling rate can be lowered, which kicks in oversampling and averaging. Configuring each ADS1220 to 175 SPS, a 3-minute recording yielded a Bluetooth sampling rate of 171 Hz, 0% packet loss, a maximum error of $\pm$ 0.05 N, and a standard deviation error of $\pm$ 0.01 N for the front force sensor (the other force sensor errors were identical or lower).

When continuously streaming over Bluetooth at its max sampling rate (ASD1220's configured to 2 kSPS), the EGG's 600 mAh battery provides a minimum of 3 hours of battery life. For additional details on Bluetooth sampling rates, packet loss %, center of mass, coefficient of friction, and both static and dynamic force errors at different test locations and force ranges, refer to our supplementary data.

D. EGG Transfer Task and Operating Modes

The EGG assesses hand function through a simple transfer task. Here, participants use one hand to quickly transfer the EGG over a 2-inch-tall barrier from one magnet to another (Fig. 1A). After each transfer, users fully release the EGG, and then retransfer the EGG in the opposite direction. As users complete these transfers, the EGG app calculates and displays quantitative performance metrics for each individual transfer completed (e.g., transfer duration, peak grip force used during the transfer, etc.) as well as summarizes data for all completed transfers (e.g., average transfer duration, average peak grip force, total number of transfers completed, etc.), thus evaluating hand performance. These transfers can be further customized via the EGG's three unique operating modes (Fig. 2B), which attempt to emphasize different aspects of sensorimotor hand function.

In "Non-Fragile" mode, the EGG is silent, and users may use any grip force they need to rapidly transfer the EGG over the barrier onto the opposing magnet. Similar to the Box and Blocks test, this mode is designed to measure gross motor hand function by focusing primarily on object transfer speed, and it does not explicitly require fine motor control.

In "Fragile" mode, we introduce a "break" sound that triggers if the grip force applied to the EGG exceeds a set break threshold. This mode demands both speed and that the user avoid breaking the object, thereby placing a greater emphasis on fine motor control and integration of tactile sensory feedback. The break threshold is fully programmable and is set directly within the EGG app.

In "Fragile-Feedback" mode, assistive audiovisual feedback is added to fragile transfers via a square-wave sound (which increases in frequency from 100 to 1000 Hz as applied grip force increases), and an LED light bar (these LEDs gradually turn on as grip force increases and shift color from green to yellow to red). Upon breaking (grip force exceeding the break threshold), all LEDs flash white, and a break sound plays. This assistive feedback mode is designed to help aid individuals,

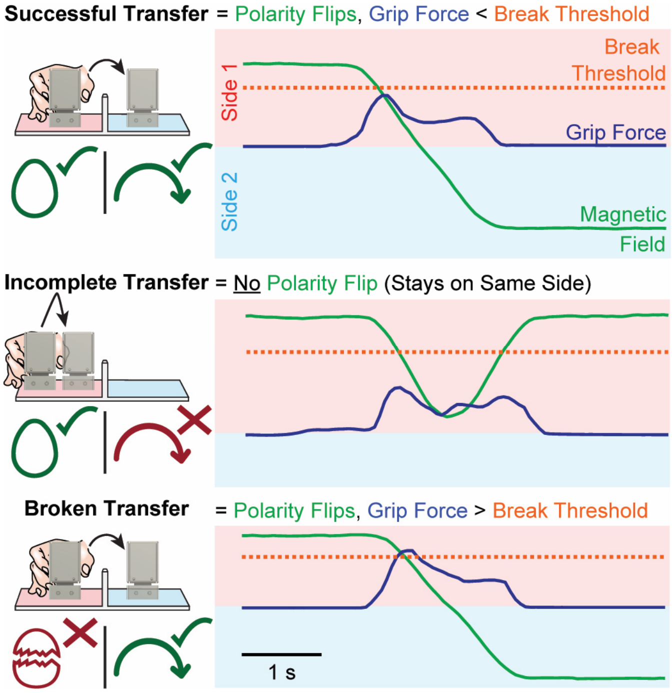

Fig. 3. Automatic Transfer Labeling. During a grasping window, grip force data (blue) and magnetic field data (green) are used to label the type of EGG transfer performed. Here we showcase example data captured from the EGG during fragile mode and their accompanying transfer labels. During a successful transfer, the EGG is moved from one magnet to another, flipping the polarity of the magnetic field, and the grip force stays below the break threshold. During an incomplete transfer, the EGG stays on the same side, and the magnetic field does not switch polarity. During a broken transfer the polarity flips (indicating the EGG has been moved to an opposing magnet), but the applied grip force exceeds the break threshold. During a broken and incomplete transfer both criteria are met with the applied grip force exceeding the break threshold and the polarity never flipping. If multiple transfers are completed within a single grasp (e.g., not letting go), an Unreleased Transfer label is assigned. During non-fragile mode, broken labels are not given.

Fig. 4. Screenshots of the EGG App running on a phone, with a Transfer History chart visible (y-axis = grip force, x-axis = datapoint index). This chart displays previously completed grasps (red) and ongoing grasps (blue). Here the EGG was set to fragile-feedback mode, with a break threshold of 8 N (dotted red line). In the History tab, individual performance metrics for completed grasps appear alongside table filtering options. The Summary tab displays transfer counts and a table which can be filtered to average the first N, the last N, or all transfers. The charts tab lets the user display other sensor data. Bottom app buttons allow the user to clear all grasp data, start a recording, tare the force sensors, and look up results from prior recordings.

especially those with tactile sensory feedback issues, and is intended to be used as a training and/or sensorimotor re-education tool. All sounds generated come from the EGGs connected computing device, which is placed near the participant. Alternative feedback sounds such as a metronome-like ticking sound, etc., can also be selected through the app.

E. Automatic Performance Metrics

We have developed a relative position tracking system for the EGG that utilizes a Hall-effect sensor and two magnets of opposing polarity. The Hall-effect sensor measures magnetic field strength and polarity, enabling the EGG to detect when it is resting on a negative or positive magnet (magnetic field polarity) and determine the vertical height above a magnet (magnitude of magnetic field strength). Coupled with grip force tracking, this position tracking system allows us to fully automate data analysis, allowing the EGG to determine when and what type of transfer has occurred (e.g., Successful Transfer, Incomplete Transfer, Broken Transfer) (Fig. 3) and then convert data within individual grasping windows into useful performance metrics (e.g., transfer time, peak grip force, EGG break status etc.).

A grasping window begins when the applied grip force exceeds 0.5 N and ends when it falls back below 0.5 N. Within a grasping window, the EGG is said to have been moved from one magnet to another if the polarity flips. A polarity flip occurs if the difference between the maximum and minimum magnetic field strength measured by the Hall-effect sensor exceeds 20 mT within the grasping window.

This entire system runs in real time via our custom-made app running on the computing device (cell phone, desktop, etc), automatically labelling transfers and generating their associated performance metrics immediately after the user finishes a grasp (Fig. 4). This app also has a recording button, which when active, continuously saves EGG data into a singular SQLite database file for later offline analysis or future display within the app, allowing users to track long term performance. This database file packages data into tables and can be opened by programs such as Excel and MATLAB. Data saved includes the transfer labels for each grasp completed and their associated performance metrics, raw sensor data, and timestamps.

III. Methods

A. Participants and Screening Assessments

Twenty-six individuals (ages 23-78, mean age 50.1 years, 50% male) with upper-limb hemiparesis due to prior stroke (N = 21), traumatic brain injury (N = 2), or cerebral palsy (N = 3) participated in this study. Among those with stroke or TBI (n = 23), 78.2% (18/23) were right-hand dominant before their injury, and for 47.8% (11/23), their paretic limb was previously their dominant hand. Prior to recruitment, all hemiparetic participants had to meet the following eligibility criteria: ability to grasp, transfer, and release the EGG using their paretic hand without assistance.

In addition to this hemiparetic group, an age-matched and gender balanced healthy control group consisting of 26 individuals (ages 23-78, mean age 50.1 years, 50% male) with no known hand deficits participated in this study, of whom 96% (25/26) were right-hand dominant. Informed consent and experimental protocols were carried out in accordance with the University of Utah Institutional Review Board (IRB #00098851) and Declaration of Helsinki.

Prior to using the EGG, the hemiparetic group's hand function was screened to ensure deficits were present in the paretic hand and that the participants recruited were not fully recovered. Here, participants were screened using a self-assessment survey and a series of standardized clinical tests. These tests evaluated motor and tactile sensory function, grip and pinch strength, spasticity, and proprioception.

More specifically, participants self-rated their hand's ability to perceive touch and its motor function on a 0 (no function) to 10 (normal) scale. Motor function was then measured using the BBT and PPT, with higher scores indicating better hand function [13], [15], and the 9HPT, with lower completion times indicating better function [14]. In the PPT, only the portion that timed the number of pegs placed with one hand in 30 seconds was tested. Sensory function was then assessed for each finger using the 2PDT for spatial acuity (scores of 0 indicating anesthesia and 4 indicating normal) [38] and the SWMT for the ability to perceive touch (0 = anesthesia, 5 = normal) [39]. Max pinch strength was assessed using a key grip and Jamar pinch gauge, while max grip strength was evaluated using a Jamar dynamometer. Spasticity was quantified at the MCP joints (knuckles) using the Modified Ashworth Scale (MAS) (0 = no rigidity, 5 = full rigidity) [40]. Finally, proprioception was tested at the thumb using the Fugl-Meyer upper extremity tests guidelines (0 = no proprioception, 2 = normal) [41]. Summarized results from these preliminary screenings are available as supplementary data (Table S2).

From these preliminary screenings, we found that in the hemiparetic group, in addition to having motor impairments, 12 out of 26 paretic hands also presented with sensory impairments as defined by a SWMT score of ≤ 4 on at least one finger. Within these 12 participants, 4 had complete anesthesia (SWMT = 0) in at least one digit on their paretic hand.

B. The EGG Transfer Task

After screening hand function using self-ratings and clinical tests, all participants then engaged in the EGG transfer task. Here, participants were instructed to transfer the EGG over a 2-inch-tall 3D-printed barrier as quickly as possible, 20 times per hand, under three distinct EGG modes: Non-Fragile, then Fragile and finally Fragile-Feedback in that order (refer to Fig. 2). We did not randomize the order of the conditions and instead chose to always perform the conditions in the same order. The Non-Fragile task was always performed first to avoid biasing the participant's naturally self-selected grip force. Similarly, the Fragile task was performed before the Fragile-Feedback task to avoid biasing the participant's fine grip force. The Fragile task provides only binary information when the applied force is too high; completing this task first allows for a more thorough and unbiased view of their default grip force regulation in the absence of the more granular feedback that may be more optimal for learning.

During Fragile and Fragile-Feedback modes participants were instructed to transfer the object as quickly as possible without breaking it and were shown demonstrations on how the mode operated (e.g., the break sound and the audio-visual feedback). For each mode, the non-paretic (paretic group) or non-dominant (control group) hand completed its 20 transfers first and consecutively, before swapping to the other hand. Due to the high number of repeated transfers across both hands and the simplicity of the task, no familiarization periods were given. To prevent potential fatigue, participants were allowed to rest in between transfers. Since the EGG metrics use only data collected during transfers, rests in between transfers did not alter the results recorded.

Participants fully released the EGG after each transfer before repeating the transfer in the opposite direction using the same hand. For each transfer completed, data on applied grip force, vertical load force, and transfer duration were recorded for analysis. Analysis was restricted to the 20 completed transfers per test condition (consisting of both broken and successful transfers). Incomplete attempts (EGG grasped but not transferred) and unreleased transfers (accidentally completing multiple transfers without letting go) were omitted. Although not analyzed in this present work, data were also collected on participants X, Y, and Z accelerations, which may have value in future work involving hand tremor detection [42].

In this study, the EGG weighed 296 grams and had a programmed break threshold of 8 Newtons (N), such that the ratio of break force to weight used was 0.027 N/g, as used in our previous study [20]. Other studies have used ratios of 0.032 N/G, 0.02 N/g, 0.015 N/g and 1.34 N/g [21], [24], [25], [27], [43]. During Fragile-Feedback mode, participants used a square-wave sound for feedback, which varied in frequency from 100 to 1000 Hz linearly and proportionally to applied grip force, in addition to the proportional visual feedback provided by the LEDs that was previously described. During data collection, EGG data was streamed at roughly 100 Hz from the EGG to a computing device (laptop) in the vicinity of the participant.

C. EGG Metrics Collected

Each participant completed 20 EGG transfers per hand per mode. For each participant's 20 EGG transfers per hand per mode, we then calculated the following Individual-level performance metrics: average transfer duration, average peak grip force, and number of EGGs broken out of 20 (if applicable to the mode).

In addition to these metrics, we also looked at the variability in the applied grip force over time, using a new metric we are calling force variability. Using each participant's

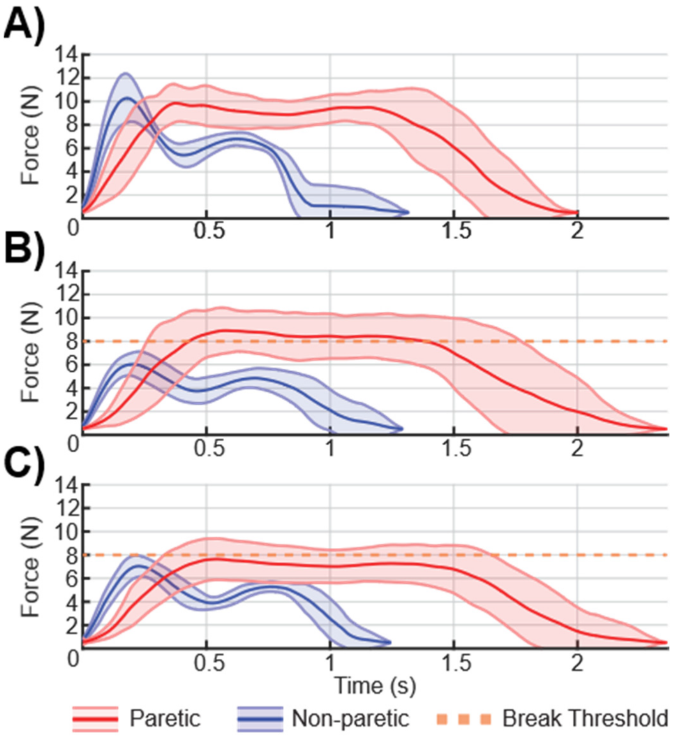

Fig. 5. Example grasping forces for A) Non-fragile, B) Fragile, and C) Fragile-Feedback transfers, from a single hemiparetic stroke participant. Data shows grip force mean ± standard deviation for the paretic (red) and non-paretic (blue) hand. N = 20 transfers per hand per mode. Note how the participants paretic hand struggles to minimize grip force during fragile transfers with the average grip force exceeding the 8 N break threshold. Additionally, notice how the average grip force of the paretic hand decreases below 8 N when given assistive feedback (fragile-feedback mode) with no substantial change in transfer speed.

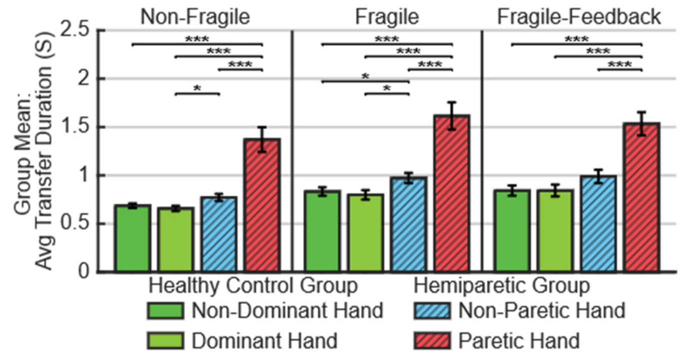

Fig. 6. Group Mean: Average EGG transfer duration across EGG modes and hands. Bars represent the group mean of the individual-level average transfer durations ± standard error of the mean (SEM). n = 26 healthy control participants and n = 26 hemiparetic participants. Significant difference markers: *** = $p < .001$, ** = $p < .01$, * = $p < .05$. For the stats tests used, refer to the Methods section D.

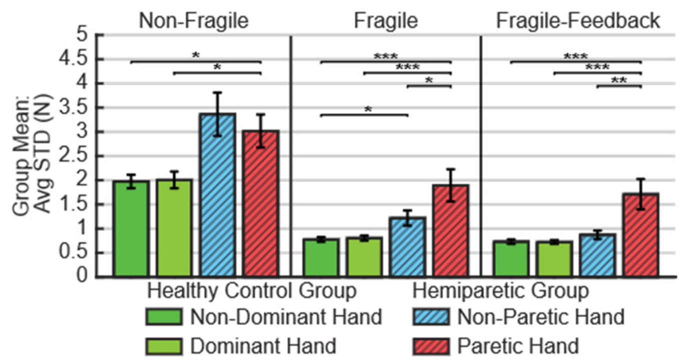

Fig. 7. Group Mean: Variability in applied grip force over time (Average STD) across EGG modes and hands. Bars represent the group mean of the individual-level average STDs ± SEM. n = 26 healthy control participants and n = 26 hemiparetic participants. Significant difference markers: *** = $p < .001$, ** = $p < .01$, * = $p < .05$. See Methods section D for stats used.

20 transfers per condition, we calculated the standard deviation of the applied grip force at each point in time. We then took the average of these standard deviations to get the force variability. This metric looks at the similarity in applied grip force between transfers, a force variability of zero indicating that all 20 transfers are identical in grip forces used at each point in time.

Finally, we also calculated some additional metrics that are showcased in the supplemental data section (Fig. S2 & Fig. S3). Average Grasp to Liftoff Delay is a metric that looks at the average time delay between the EGG being grasped and lifted vertically off the magnet [29]. Average transfer grip force is the average grip force used over the course of an entire transfer.

D. Statistical Analysis

1) Group-Level: After collecting these individual-level performance metrics (e.g., average transfer duration, average peak grip force, EGGs broken out at 20, force variability, average grasp-to-liftoff delay, and average transfer force) for each individual participant, we then averaged the individual-level performance metrics together into group-level means (e.g., group mean: average transfer duration, group mean: average peak grip force, etc.). Here, group means were calculated for the following four groups: healthy control participants' non-dominant and dominant hands, hemiparetic participants' non-paretic and paretic hands. To determine whether there was a statistically significant difference between group means ($p < 0.05$), nonparametric statistical tests were utilized and corrected with Benjamini-Hochberg (BH) to avoid type 1 error via multiple comparisons. Here we used Wilcoxon signed-rank tests for paired data (e.g., same subject, left vs. right hand) and Wilcoxon rank-sum tests for unpaired data (e.g., hemiparetic vs. healthy control group) (Figs. 6–9). Comparisons between EGG modes utilized Wilcoxon Signed Rank with BH and are available as supplementary data (Figs. S4-S9).

2) Individual-Level: In addition to comparing group-level means, we also ran statistics comparing individual-level metrics (e.g., average transfer duration, etc.). Specifically, we looked at our hemiparetic group and quantified the number of individuals who saw a statistically significant ($p < 0.05$) decrease or increase in EGG performance between fragile and fragile-feedback modes when utilizing the paretic or non-paretic hand (Table I). This individual-level analysis relied on comparing the EGG metric data collected from an individual's 20 transfers per test condition (e.g., comparing data from participants 20 Fragile vs 20 Fragile-Feedback transfers). Each transfer was treated as an independent observation (order

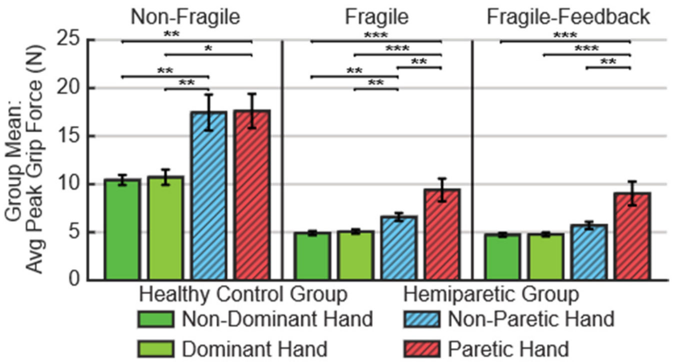

Fig. 8. Group Mean: Average peak grip force used across modes and hands (bars represent the group mean ± SEM). n = 26 healthy control participants and n = 26 hemiparetic participants. Significant difference markers: *** = $p < .001$, ** = $p < .01$, * = $p < .05$. See Methods section D for stats used.

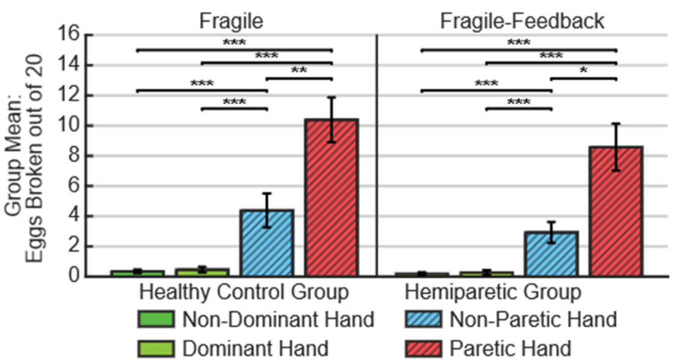

Fig. 9. Group Mean: EGGs broken out of 20 across fragile and fragile-feedback modes (bars represent the group mean ± SEM). Significant difference markers: *** = $p < .001$, ** = $p < .01$, * = $p < .05$. See Methods section D for stats used.

TABLE I

HEMIPARETIC HANDS: NUMBER WHO IMPROVED, WORSENED, OR DID NOT CHANGE, AFTER FRAGILE EGG TRANSFERS WERE GIVEN AUDIOVISUAL FEEDBACK

	EGGs broken out of 20		Avg Peak Grip Force		Avg Transfer Duration		Force Variability	
Hand	Non-Par	Par	Non-Par	Par	Non-Par	Par	Non-Par	Par
# Improved	4	6	14	10	8	10	15	18
# Worsened	0	2	4	5	8	7	4	5
# No change	22	18	8	11	10	9	7	3

Note: Par is short for paretic. N = 26 hemiparetic participants. "# improved" denotes the number of participants with a statistically significant improvement (fewer eggs broken, lower peak grip force, shorter transfer duration, lower force variability) ($p<0.05$) when fragile transfers were given feedback.

assumed not to matter). For continuous data (average transfer duration, average peak grip force, force variability), we used the Wilcoxon Rank Sum. For binary data (EGGs broken out of 20 transfers), we used Fisher's exact test.

IV. RESULTS

A. The EGG Is Capable of Capturing Fundamental Information on Sensorimotor Hand Function

Data from the EGG were used to generate plots of applied grip force over time. These plots highlight key differences between a participant's paretic and non-paretic limb. For example, here we showcase EGG transfer data from one of the hemiparetic patients across different EGG modes (Fig. 5). This participant's data is presented as a case study due to the differences between their non-paretic and paretic hand being fairly representative of the entire hemiparetic cohort.

Compared to the non-paretic hand, this participant's paretic hand had a higher degree of force variability (larger average standard deviation) during Fragile and Fragile-Feedback modes. Meaning that while the non-paretic hand tended to transfer the EGG with a consistent grip force and speed, the paretic hand transfers were less consistent and had more variation in grip force used at each point in time from transfer to transfer.

Additionally, this participant's paretic hand transferred the EGG more slowly, struggled to minimize grip force during fragile transfers, and ultimately broke the EGG more often than the non-paretic hand. When assistive feedback was enabled, this particular participant's average peak grip force saw a statistically significant decrease from 9.7 N to 8.1 N ($p < 0.05$, Wilcoxon Rank Sum, effect size r = 0.40 Rosenthal correlation coefficient [44]). They also went from breaking 16 EGGs during fragile mode to 11 during Fragile-Feedback mode; however, this change was not statistically significant ($p < 0.05$, Fisher's exact test).

B. On Average, Paretic Hands Transfer the EGG Significantly Slower Than Non-Paretic and Healthy Control Hands

Across each EGG mode tested (Non-Fragile, Fragile, and Fragile-Feedback), a clear performance hierarchy emerged. Paretic hands transferred the EGG significantly slower than non-paretic hands and both healthy control hands (Fig. 6). Notably, within each mode tested, the non-dominant and dominant healthy control hands performed at roughly the same speed.

In our supplemental data, the average grip force load force delay showed similar findings. On average, across all modes, paretic hands took a significantly longer time grasping the EGG before they were able to lift it off the magnet (Fig. S2).

C. Paretic Hands Had Significantly More Variability in Applied Grip Force During Fragile and Fragile Feedback Transfers Compared to Healthy and Non-paretic Hands

During Fragile and Fragile-Feedback transfers, the applied grip forces of the hemiparetic groups' paretic hands had significantly more force variability than the healthy control groups' hands and the hemiparetic groups' non-paretic hands. In essence, the healthy control group and non-paretic hands tended to consistently transfer the EGG with the same amount of grip force and speed across all 20 transfers performed per mode per hand. Meanwhile, the hemiparetic group's paretic hands were less consistent, tending to have more variation in applied grip force at each time point, transfer to transfer (Fig. 7).

D. Paretic Hands Struggle to Minimize Grip Force During Fragile EGG Transfers

During both Fragile and Fragile-Feedback modes, the average peak grip force generated by the paretic hand was significantly higher than the non-paretic and healthy control hands (Fig. 8), indicating difficulty with grip force minimization and resulting in breaking the EGG more often.

Additionally, the average peak grip force of the non-paretic hand was also found to be higher than the healthy control group's hands during Fragile mode, resulting in more EGGs breaking. Although hemiparesis affects one side more strongly, these results showcase how the "non-paretic" side can also experience minor deficits, as seen in prior literature [45].

E. Paretic Hands Broke the EGG More Often Than Non-Paretic and Healthy Control Hands

Across both Fragile and Fragile-Feedback conditions, on average, paretic hands broke the EGG significantly more often than both non-paretic and healthy control hands (Fig. 9). Healthy control hands performed the best, with no significant differences between non-dominant and dominant control hands. When comparing group means across modes, enabling assistive audiovisual feedback (Fragile-Feedback mode) did not make a statistically significant difference on the number of EGGs broken by each hand tested (Fig. S7).

F. For a Subset of Hemiparetic Participants, Enabling Audiovisual Feedback Significantly Improved Paretic Hand Performance

For paretic hands, when comparing group means across Fragile and Fragile-Feedback modes, with the exception of force variability, enabling assistive audiovisual feedback did not have a statistically significant impact on the group means for number of eggs broken, avg peak grip force and avg transfer duration (Figs. S4-S7). However, during individual statistical analysis, we found that a subset of participants' paretic hands saw a statistically significant improvement (Table. I). Notably, enabling feedback generally resulted in participants' performance either not changing or improving, with only a minority worsening when feedback was enabled.

Complimenting these aggregate findings (Table. I), we also provide data on a per-participant basis. More specifically, we calculated the cumulative % change in each participant's performance metrics after feedback was enabled for the paretic hand (Fig. 10).

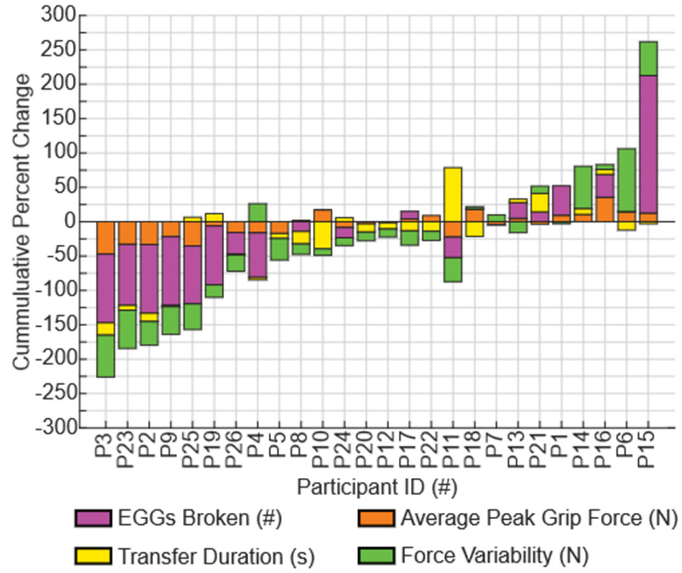

Fig. 10. Cumulative percent change in EGG performance metrics for the paretic hand after assistive audiovisual feedback was enabled. N = 26 participants (P#). Percent Change = 100*(Fragile-Feedback Mode–Fragile Mode) / Fragile Mode, for each specific EGG performance metrics. Y axis shows the cumulative percent change as bars with the same sign (±) are stacked on top of each other when in the same direction. Data is ordered from largest improvement to worst improvement. For example, P3 saw the largest cumulative improvement after feedback was enabled with all four metrics decreasing (EGGs broken, Average Peak Grip Force, Transfer Duration, Force Variability), for a total cumulative change of -226%. Note that % change in EGGs broken was omitted for P5, P7, P14, and P22, as they broke zero eggs in Fragile mode, making calculation of % change impossible due to division by zero. However, these four participants also broke zero eggs during the Fragile-Feedback mode making their performance change functionally zero.

V. Discussion

In this study, we introduced the EGG, a novel instrumented device designed to create automatic, high-resolution performance metrics for sensorimotor hand function. Utilizing the EGG's automated performance metrics, we then evaluated the hand function of 26 hemiparetic and 26 healthy age-matched individuals as they completed transfers across three distinct modes (Non-Fragile, Fragile, and Fragile-Feedback), providing a novel comparative analysis of hemiparetic hand function and showcasing the EGG's real-world utility.

To our knowledge, this paper is among the first to use instrumented fragile object transfers to directly quantify differences in grip force regulation and transfer speed between hemiparetic participants' paretic and non-paretic hands at this scale and with accompanying healthy age-matched controls' dominant and non-dominant hands. This detailed quantitative analysis was enabled by the EGG's unique electronic design. In the context of prior works for evaluating grip force control, the EGG introduces significant advancements and addresses many prior limitations associated with both analog and electronic devices. The closest analog approximations to the EGG transfer task are the Strength-Dexterity (S-D) test [46], which uses compressible springs, and the Virtual Eggs Test (VET) [35], which uses transfers of mechanical blocks that collapse at known force thresholds. Meanwhile, the closest electronic approximations to EGG are commercial digital dynamometers such as the GripAble [47] and Squegg [48], as well as instrumented devices (e.g., handles and cubes) used primarily in laboratories [29], [30], [32].

Compared with digital dynamometers and instrumented laboratory devices, the EGG offers several key innovations: namely, its focus on evaluating grip force control during rapid, fragile-object transfers; its fully automated performance-tracking system; and the integration of assistive audio-visual feedback. Many prior works and devices have predominantly focused on evaluating grip force control under static conditions, such as grip–lift–hold tasks in which participants attempt to hold an object stationary in the air while keeping the applied force below a prescribed threshold [19], [49], [50], or force-tracking tasks in which the device is held still while users visually track their applied force on a computer display [51], [52]. In contrast, the EGG transfer task emphasizes force regulation during active object motion. Because many activities of daily living require precise grip force control while an object is being transported, the EGG offers a unique opportunity to both evaluate dynamic grip force regulation and potentially serve as a training tool for practicing grip force control under the guidance of assistive feedback.

On the technical front, the EGG offers improvements in general portability over many instrument laboratory devices which may be tethered to external power sources, computer displays, or data loggers [19]. The EGG is fully wireless and data can be collected from a cellphone. Critically, the EGG can achieve Bluetooth sampling rates of over 900 Hz, with packet losses below 3%, and average maximum dynamic force errors of $\pm$ 0.28 N. At lower, more clinically relevant, Bluetooth sampling rates (e.g., 171 Hz), packet losses are 0%, and average maximum dynamic force errors are $\pm$ 0.16 N. Supplementary Tables S5 & S6 provide additional information regarding dynamic force errors and Bluetooth sampling rates.

Additionally, a limitation of many analog systems is their reliance on manual data collection. For example, to calculate object transfer durations, prior studies have manually timed the start and end of each object transferred using a stopwatch [24], [53]. Manually timing transfers introduces human error and can be both tedious and inconvenient. Furthermore, the potential for timing inaccuracies can be pronounced among individuals with hemiparesis. Paretic hands often struggle with both the initial grasp and the final release of objects [54]. Without sensors that directly relay grasping forces, it can be difficult to determine when a grasp starts and ends [33]. The EGG utilizes a unique relative position tracking system, which, in combination with measuring applied grip force, allows us to label the type of transfer that occurred and then immediately and automatically provide performance metrics [55]. These performance metrics are then displayed via a custom app, with an option to save and export data for future offline analysis or long-term performance tracking. These features represent a benefit over methods that require manual data collection.

In addition to these advancements, the EGG introduces three novel operating modes designed with specific goals in mind: Non-Fragile mode for evaluating gross motor function, Fragile mode for evaluating fine sensorimotor function, and Fragile-Feedback mode as a rehabilitative tool for helping participants practice and potentially reeducate fine sensorimotor function. Within this study we utilized these three different modes to evaluate the hand function of hemiparetic and healthy individuals as they completed EGG transfers.

Compared to non-paretic and healthy control hands, paretic hands were not only significantly slower when performing the EGG transfer task across all modes, but also showed profound difficulty regulating grip force, resulting in higher average peak forces and more EGGs broken during fragile object transfers. Notably, our findings dissociate the ability to maximize grip force (strength) from the ability to regulate it (control). In the hemiparetic group tested: compared to the non-paretic hand, paretic hands on average exhibited weaker maximum grip strength (paretic: 16.5 $\pm$ 9.6 kg max grip force vs non-paretic: 34.6 $\pm$ 10.8 kg max grip force; Table S2), while applying significantly higher peak grip forces during fragile mode EGG transfers (paretic: 9.4 $\pm$ 1.2 N vs non-paretic: 6.6 $\pm$ 2.1 N; Fig. 8 & Table S9) and consequently breaking the EGG more often during fragile transfers (paretic: 10.4 $\pm$ 1.5 EGGs broken vs nonparetic: 4.4 $\pm$ 1.1 EGGs broken; Fig. 9 & Table S9).

If weak grip strength alone were the issue, one might expect a struggle to apply sufficient force and fewer EGGs being broken; instead, many of these participants' paretic hands consistently applied excessive force during fragile transfers. These findings suggest that difficulty in controlling applied grip force, rather than lack of grip strength, was a primary factor driving the number of EGGs broken during fragile transfers by the paretic hand. This observation is congruent with prior literature which looked at the applied grip forces of hemiparetic stroke patients when asked to hold a small object stationary in the air. These studies found that participants' paretic hands, despite having weaker maximum grip forces (i.e., less strength), frequently applied grasping forces that were excessive and significantly greater than the amount needed to prevent object slippage. Paretic grasping forces were also often significantly higher than both the non-paretic hand and healthy control hands [56], [57], [58].

The impact of adding audiovisual feedback further highlights the complexity of control deficits in the paretic hand. When comparing group means for paretic hand (Figs. S4, S6, & S7), feedback did not produce a statistically significant improvement in the group means for average transfer time, average peak force, or EGGs broken. However, a closer look at individual-level performance revealed a subset of hemiparetic participants whose paretic hands benefited significantly from the feedback, with 10/26 reducing their average peak grip force, 6/26 breaking significantly fewer EGGs out of 20, and 10/26 transferring the EGG faster when feedback was enabled (Table I). Interestingly, enabling feedback for the paretic hand almost always improved or maintained performance, with only a minority doing significantly worse (5/26 had higher peak grip forces, 2/26 broke the egg more often, 7/26 transferred the EGG more slowly).

Notably, in the six individuals who broke significantly fewer EGGs when given feedback we did not find sensory deficits (SWMT at the thumb) that were significantly different ($p < 0.05$, Wilcoxon rank-sum) from the overall cohort mean. These findings suggest that sensory function alone was not predictive of whether assistive feedback would improve performance. In reality, whether feedback helps improve performance is likely influenced by a complex integration of several factors, such as motor function, sensory function, spasticity, and strength.

We speculate that there may exist a "sweet spot" in paretic hand function where audiovisual feedback can help improve performance. Participants who have very little hand function may be struggling to utilize the additional feedback, resulting in no improvement, while on the opposite spectrum, those with high levels of function (minor paresis) may already be performing near-perfectly, gaining little benefit.

Methodologically, we note that although participants received a demonstration of the EGG transfer task and its modes, no practice period was provided, and the testing order remained fixed. Given the task's simplicity and the high volume of transfers performed, we deemed additional familiarization unnecessary. This approach aligns with similar object transfer assessments, such as the validated BBT, where practice is minimal and optional. On average, healthy participants showed no significant performance differences between hands, despite the dominant hand being tested second and theoretically benefiting from practice with the non-dominant hand. Furthermore, the high volume of transfers inherently dilutes the impact of a potential early learning effect. The paretic hand was also tested second and completed Non-Fragile transfers prior to the Fragile modes, ensuring inherent familiarity with the EGG before more demanding tasks were assessed. Nevertheless, we cannot completely rule out learning and order effects as covariates in this study. Future studies could explore randomizing testing orders to further minimize order effects.

Although this work has laid the foundation by establishing the EGGs design, technical innovations, and preliminary clinical utility, further research is required to validate the device and its performance metrics. Future work should focus on determining the correlation between the EGG metrics and other standardized clinical assessments (concurrent validity) and investigating other psychometric properties, such as test-retest reliability and minimum detectable change, within a single patient population (e.g., stroke without TBI or CP patients).

VI. Conclusion

This study introduced the EGG, a novel instrumented fragile object that provides automatic, high-resolution metrics for gross motor and fine sensorimotor hand function. Using this device, we identified key differences in hand function between individuals with upper-limb hemiparesis and healthy controls. Notably, the EGGs quantitative metrics captured deficits in the paretic hand's ability to precisely regulate grip force, a dimension of hand function frequently overlooked by standard clinical assessments. We also demonstrated that audiovisual feedback could help improve grip force control for a subset of hemiparetic participants, highlighting the EGG's dual potential as both an assessment and targeted neurorehabilitation tool. Ultimately, this work helps support the use of instrumented fragile objects as tools for providing objective, high-resolution data on hand dexterity, paving the way for next-generation diagnostic and rehabilitative strategies in neurorehabilitation. In an effort to translate this device from the lab and into clinic, those interested in using the EGG can learn more at www.egg.rehab.

Author's Contributions

Michael D. Adkins designed, developed and built both the EGG and its app, recruited participants, collected and analyzed data, and wrote this manuscript. Tyler S. Davis developed an early MATLAB version of the EGG app which helped pilot test features. Tyler J. Gourley assisted with revision data collection. Nate Toth assisted in designing an early version of the EGGs housing. Monika K. Buczak assisted with data collection. Steven R. Edgley provided access to patients and clinical insight. Masaru Teramoto assisted with statistics. Lorie G. Richards provided device feedback. Marta M. Iversen and Jacob A. George helped supervise and guide the study. All authors reviewed and edited the manuscript.

Acknowledgment

The authors would like to thank all the participants who participated in this study, without whom the testing and development of the EGG device would not have been possible.

References

[1] S. M. Hatem et al., "Rehabilitation of motor function after stroke: A multiple systematic review focused on techniques to stimulate upper extremity recovery," *Frontiers Human Neurosci.*, vol. 10, p. 442, Sep. 2016, doi: 10.3389/fnhum.2016.00442.

[2] D. I. Katz, M. P. Alexander, and R. B. Klein, "Recovery of arm function in patients with paresis after traumatic brain injury," *Arch. Phys. Med. Rehabil.*, vol. 79, no. 5, pp. 488–493, May 1998, doi: 10.1016/s0003-9993(98)90060-0.

[3] P. Uvebrant, "Hemiplegic cerebral palsy aetiology and outcome," *Acta Paediatrica*, vol. 77, pp. 1–100, May 1988, doi: 10.1111/j.1651-2227.1988.tb14939.x.

[4] C. E. Lang, M. D. Bland, R. R. Bailey, S. Y. Schaefer, and R. L. Birkenmeier, "Assessment of upper extremity impairment, function, and activity after stroke: Foundations for clinical decision making," *J. Hand Therapy*, vol. 26, no. 2, pp. 104–115, Apr. 2013, doi: 10.1016/j.jht.2012.06.005.

[5] S. Doyle, S. Bennett, S. E. Fasoli, and K. T. McKenna, "Interventions for sensory impairment in the upper limb after stroke," *Cochrane Database Systematic Rev.*, vol. 2016, no. 6, Jun. 2010, Art. no. 006331, doi: 10.1002/14651858.cd006331.pub2.

[6] L. Poltawski et al., "Assessing the impact of upper limb disability following stroke: A qualitative enquiry using internet-based personal accounts of stroke survivors," *Disability Rehabil.*, vol. 38, no. 10, pp. 945–951, May 2016, doi: 10.3109/09638288.2015.1068383.

[7] T. L. T. Lo, J. L. C. Lee, and R. T. H. Ho, "Recovery beyond functional restoration: A systematic review of qualitative studies of the embodied experiences of people who have survived a stroke," *BMJ Open*, vol. 13, no. 2, Feb. 2023, Art. no. e066597, doi: 10.1136/bmjopen-2022-066597.

[8] J. M. Cassidy and S. C. Cramer, "Spontaneous and therapeutic-induced mechanisms of functional recovery after stroke," *Transl. Stroke Res.*, vol. 8, no. 1, pp. 33–46, Feb. 2017, doi: 10.1007/s12975-016-0467-5.

[9] L. A. Boyd et al., "Biomarkers of stroke recovery: Consensus-based core recommendations from the stroke recovery and rehabilitation roundtable," *Int. J. Stroke*, vol. 12, no. 5, pp. 480–493, Jul. 2017, doi: 10.1177/1747493017714176.

[10] K. Dworzynski, G. Ritchie, and E. D. Playford, "Stroke rehabilitation: Long-term rehabilitation after stroke," *Clin. Med.*, vol. 15, no. 5, pp. 461–464, Oct. 2015, doi: 10.7861/clinmedicine.15-5-461.

[11] L. Edwards, E. M. King, C. M. Buetefisch, and M. R. Borich, "Putting the 'sensory' into sensorimotor control: The role of sensorimotor integration in goal-directed hand movements after stroke," *Frontiers Integr. Neurosci.*, vol. 13, p. 16, Aug. 2019. [Online]. Available: https://www.frontiersin.org/articles/10.3389/fnint.2019.00016

[12] N. Bolognini, C. Russo, and D. J. Edwards, "The sensory side of post-stroke motor rehabilitation," *Restorative Neurol. Neurosci.*, vol. 34, no. 4, pp. 571–586, Aug. 2016, doi: 10.3233/rnn-150606.

[13] V. Mathiowetz, G. Volland, N. Kashman, and K. Weber, "Adult norms for the box and block test of manual dexterity," *Amer. J. Occupational Therapy*, vol. 39, no. 6, pp. 386–391, Jun. 1985, doi: 10.5014/ajot.39.6.386.

[14] K. Oxford Grice, K. A. Vogel, V. Le, A. Mitchell, S. Muniz, and M. A. Vollmer, "Adult norms for a commercially available nine hole peg test for finger dexterity," *Amer. J. Occupational Therapy*, vol. 57, no. 5, pp. 570–573, Sep. 2003, doi: 10.5014/ajot.57.5.570.

[15] M. Stijic, K. Petrovic, P. Schwingenschuh, M. Koini, and R. Schmidt, "The Purdue pegboard test: Normative data from 1,355 healthy people from Austria," *Amer. J. Occupational Therapy*, vol. 77, no. 3, May 2023, Art. no. 7703205030, doi: 10.5014/ajot.2023.050023.

[16] D. S. Louis, T. L. Greene, K. E. Jacobson, C. Rasmussen, P. Kolowich, and S. A. Goldstein, "Evaluation of normal values for stationary and moving two-point discrimination in the hand," *J. Hand Surg.*, vol. 9, no. 4, pp. 552–555, Jul. 1984, doi: 10.1016/s0363-5023(84)80109-4.

[17] J. Bell-Krotoski and E. Tomancik, "The repeatability of testing with Semmes–Weinstein monofilaments," *J. Hand Surg.*, vol. 12, no. 1, pp. 155–161, Jan. 1987, doi: 10.1016/s0363-5023(87)80189-2.

[18] M. Buczak, B. Baum, C. Olsen, and J. George, "Disentangling sensory and motor deficits of fine hand function using an electronic grip gauge (EGG) to simulate transferring fragile objects," presented at the *Myoelectric Controls Upper Limb Prosthetics Symp., Fredericton*, Aug. 2022, pp. 202–205.

[19] H.-Y. Hsu, C.-F. Lin, F.-C. Su, H.-T. Kuo, H.-Y. Chiu, and L.-C. Kuo, "Clinical application of computerized evaluation and re-education biofeedback prototype for sensorimotor control of the hand in stroke patients," *J. NeuroEngineering Rehabil.*, vol. 9, no. 1, p. 26, May 2012, doi: 10.1186/1743-0003-9-26.

[20] M. D. Adkins, M. K. Buczak, C. D. Olsen, M. M. Iversen, and J. A. George, "Automated quantifiable assessments of sensorimotor function using an instrumented fragile object," in *Proc. Int. Conf. Rehabil. Robot. (ICORR)*, Sep. 2023, pp. 1–6, doi: 10.1109/icorr58425.2023.10304693.

[21] F. Clemente, M. D'Alonzo, M. Controzzi, B. B. Edin, and C. Cipriani, "Non-invasive, temporally discrete feedback of object contact and release improves grasp control of closed-loop myoelectric transradial prostheses," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 24, no. 12, pp. 1314–1322, Dec. 2016, doi: 10.1109/TNSRE.2015.2500586.

[22] P. Dario et al., "Robotics as a future and emerging technology biomimetics, cybernetics, and neuro-robotics in European projects," *IEEE Robot. Autom. Mag.*, vol. 12, no. 2, pp. 29–45, Jun. 2005, doi: 10.1109/MRA.2005.1458320.

[23] E. D. Engeberg and S. Meek, "Improved grasp force sensitivity for prosthetic hands through force-derivative feedback," *IEEE Trans. Biomed. Eng.*, vol. 55, no. 2, pp. 817–821, Feb. 2008, doi: 10.1109/TBME.2007.912675.

[24] M. D. Paskett et al., "Activities of daily living with bionic arm improved by combination training and latching filter in prosthesis control comparison," *J. NeuroEngineering Rehabil.*, vol. 18, no. 1, Dec. 2021, Art. no. 1, doi: 10.1186/s12984-021-00839-x.

[25] F. M. Petrini et al., "Six-month assessment of a hand prosthesis with intraneural tactile feedback," *Ann. Neurol.*, vol. 85, no. 1, pp. 137–154, Jan. 2019, doi: 10.1002/ana.25384.

[26] G. Valle et al., "Hand control with invasive feedback is not impaired by increased cognitive load," *Frontiers Bioengineering Biotechnol.*, vol. 8, p. 287, Apr. 2020, doi: 10.3389/fbioe.2020.00287.

[27] G. Valle et al., "Biomimetic intraneural sensory feedback enhances sensation naturalness, tactile sensitivity, and manual dexterity in a bidirectional prosthesis," *Neuron*, vol. 100, no. 1, pp. 37–45.e7, Oct. 2018, doi: 10.1016/j.neuron.2018.08.033.

[28] M. Controzzi, F. Clemente, N. Pierotti, M. Bacchereti, and C. Cipriani, "Evaluation of hand function transporting fragile objects: The virtual EGGS test," in *Proc. Myoelectric Controls Up. Limb Prosthet. Symp.*, Aug. 2017, p. 1.

[29] E. Mastinu et al., "Neural feedback strategies to improve grasping coordination in neuromusculoskeletal prostheses," *Sci. Rep.*, vol. 10, no. 1, Jul. 2020, Art. no. 1, doi: 10.1038/s41598-020-67985-5.

[30] S. L. Gorniak, V. M. Zatsiorsky, and M. L. Latash, "Manipulation of a fragile object," *Exp. Brain Res.*, vol. 202, no. 2, pp. 413–430, Apr. 2010, doi: 10.1007/s00221-009-2148-z.

[31] S. L. Gorniak, V. M. Zatsiorsky, and M. L. Latash, "Manipulation of a fragile object by elderly individuals," *Exp. Brain Res.*, vol. 212, no. 4, pp. 505–516, Aug. 2011, doi: 10.1007/s00221-011-2755-3.

[32] D. A. Nowak and J. Hermsdörfer, "Grip force behavior during object manipulation in neurological disorders: Toward an objective evaluation of manual performance deficits," *Movement Disorders, Off. J. Movement Disorder Soc.*, vol. 20, no. 1, pp. 11–25, Jan. 2005, doi: 10.1002/mds.20299.

[33] M. Piazzini et al., "Application and adaptation of the virtual eggs test for assessing hand dexterity in subjects with stroke: A pilot study," *Arch. Physiotherapy*, pp. 131–137, Jun. 2025, doi: 10.33393/aop.2025.3291.

[34] D. W. Tan, M. A. Schiefer, M. W. Keith, J. R. Anderson, J. Tyler, and D. J. Tyler, "A neural interface provides long-term stable natural touch perception," *Sci. Transl. Med.*, vol. 6, no. 257, p. 257, Oct. 2014, doi: 10.1126/scitranslmed.3008669.

[35] M. Controzzi et al., "Assessing hand function in trans-radial amputees wearing myoelectric hands: The virtual eggs test (VET)," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 33, pp. 286–297, 2025, doi: 10.1109/TNSRE.2024.3524791.

[36] Y. Hiramatsu, D. Kimura, K. Kadota, T. Ito, and H. Kinoshita, "Control of precision grip force in lifting and holding of low-mass objects," *PLoS ONE*, vol. 10, no. 9, Sep. 2015, Art. no. e0138506, doi: 10.1371/journal.pone.0138506.

[37] J. A. Kleim and T. A. Jones, "Principles of experience-dependent neural plasticity: Implications for rehabilitation after brain damage," *J. Speech, Lang., Hearing Res.*, vol. 51, no. 1, pp. 225–239, Feb. 2008, doi: 10.1044/1092-4388(2008/018).

[38] P. G. Silva, A. Jones, P. M. P. Araujo, and J. Natour, "Assessment of light touch sensation in the hands of systemic sclerosis patients," *Clinics*, vol. 69, no. 9, pp. 585–588, Sep. 2014, doi: 10.6061/clinics/2014(09)02.

[39] T. A. R. Schreuders, R. W. Selles, B. T. J. van Ginneken, W. G. M. Janssen, and H. J. Stam, "Sensory evaluation of the hands in patients with charcot-marie-tooth disease using semmes-weinstein monofilaments," *J. Hand Therapy*, vol. 21, no. 1, pp. 28–34, Jan. 2008, doi: 10.1197/j.jht.2007.07.020.

[40] *(PDF) Effect of Repetitive Transcranial Magnetic Stimulation on Motor Function and Spasticity in Spastic Cerebral Palsy*. Accessed: Aug. 7, 2025. [Online]. Available: https://www.researchgate.net/publication/335638105_Effect_of_repetitive_transcranial_magnetic_stimulation_on_motor_function_and_spasticity_in_spastic_cerebral_palsy

[41] A. Fugl-Meyer, L. Jääskö, I. Leyman, S. Olsson, and S. Steglind, "The post-stroke hemiplegic patient. 1. a method for evaluation of physical performance," *J. Rehabil. Med.*, vol. 7, no. 1, pp. 13–31, Jan. 1975.

[42] K. Kitai, M. Odagiri, R. Yamauchi, and T. Kodama, "Evaluation of intervention effectiveness of sensory compensatory training with tactile discrimination feedback on sensorimotor dysfunction of the hand after stroke," *Brain Sci.*, vol. 11, no. 10, p. 1314, Oct. 2021, doi: 10.3390/brainsci11101314.

[43] J. A. George et al., "Biomimetic sensory feedback through peripheral nerve stimulation improves dexterous use of a bionic hand," *Sci. Robot.*, vol. 4, no. 32, p. 2352, Jul. 2019, doi: 10.1126/scirobotics.aax2352.

[44] R. L. Rosnow, R. Rosenthal, and D. B. Rubin, "Contrasts and correlations in effect-size estimation," *Psychol. Sci.*, vol. 11, no. 6, pp. 446–453, Nov. 2000, doi: 10.1111/1467-9280.00287.

[45] C. Maenza, D. C. Good, C. J. Winstein, D. A. Wagstaff, and R. L. Sainburg, "Functional deficits in the less-impaired arm of stroke survivors depend on hemisphere of damage and extent of paretic arm impairment," *Neurorehabilitation Neural Repair*, vol. 34, no. 1, pp. 39–50, Jan. 2020, doi: 10.1177/1545968319875951.

[46] F. J. Valero-Cuevas, N. Smaby, M. Venkadesan, M. Peterson, and T. Wright, "The strength–dexterity test as a measure of dynamic pinch performance," *J. Biomechanics*, vol. 36, no. 2, pp. 265–270, Feb. 2003, doi: 10.1016/s0021-9290(02)00340-8.

[47] R. Kudin, H. I. Ergen, and C. W. McGee, "Test–retest reliability and precision of GripAble: A multipurpose exergaming device," *Games Health J.*, vol. 14, no. 4, pp. 305–311, Aug. 2025, doi: 10.1089/g4h.2024.0216.

[48] Z. Amin, G. Gutierrez, and L. True, "Concurrent validity and test-retest reliability of Squegg-smart dynamometer and handgrip trainer in healthy individuals," *Hand Therapy*, vol. 29, no. 2, pp. 68–74, Jun. 2024, doi: 10.1177/17589983231223868.

[49] J. Monzée, Y. Lamarre, and A. M. Smith, "The effects of digital anesthesia on force control using a precision grip," *J. Neurophysiology*, vol. 89, no. 2, pp. 672–683, Feb. 2003, doi: 10.1152/jn.00434.2001.

[50] A.-S. Augurelle, A. M. Smith, T. Lejeune, and J.-L. Thonnard, "Importance of cutaneous feedback in maintaining a secure grip during manipulation of hand-held objects," *J. Neurophysiology*, vol. 89, no. 2, pp. 665–671, Feb. 2003, doi: 10.1152/jn.00249.2002.

[51] N. J. Seo, H. W. Fischer, R. A. Bogey, W. Z. Rymer, and D. G. Kamper, "Use of visual force feedback to improve digit force direction during pinch grip in persons with stroke: A pilot study," *Arch. Phys. Med. Rehabil.*, vol. 92, no. 1, pp. 24–30, Jan. 2011, doi: 10.1016/j.apmr.2010.08.016.

[52] R. Yao, Y. Morita, K. Yamada, F. Okuyama, and Y. Morita, "Impact of iWakka training on cerebral blood flow and hand function in stroke patients with left hemiplegia," in *Proc. 17th Int. Conv. Rehabil. Eng. Assistive Technol. (i-CREATe)*, Aug. 2024, pp. 1–5, doi: 10.1109/i-create62067.2024.10776224.

[53] A. L. Fougner, Ø. Stavdahl, and P. J. Kyberd, "System training and assessment in simultaneous proportional myoelectric prosthesis control," *J. NeuroEngineering Rehabil.*, vol. 11, no. 1, p. 75, Apr. 2014, doi: 10.1186/1743-0003-11-75.

[54] N. J. Seo, W. Z. Rymer, and D. G. Kamper, "Delays in grip initiation and termination in persons with stroke: Effects of arm support and active muscle stretch exercise," *J. Neurophysiology*, vol. 101, no. 6, pp. 3108–3115, Jun. 2009, doi: 10.1152/jn.91108.2008.

[55] J. A. George, "*Electric grip gauge for assessing hand dexterity*," U.S. Patent 20 240 065 603 A1, Feb. 29, 2024.

[56] J. Hermsdörfer, E. Hagl, D. A. Nowak, and C. Marquardt, "Grip force control during object manipulation in cerebral stroke," *Clin. Neurophysiol.*, vol. 114, no. 5, pp. 915–929, May 2003, doi: 10.1016/s1388-2457(03)00042-7.

[57] J. M. Blennerhassett, L. M. Carey, and T. A. Matyas, "Grip force regulation during pinch grip lifts under somatosensory guidance: Comparison between people with stroke and healthy controls," *Arch. Phys. Med. Rehabil.*, vol. 87, no. 3, pp. 418–429, Mar. 2006, doi: 10.1016/j.apmr.2005.11.018.

[58] P. G. Lindberg, N. Roche, J. Robertson, A. Roby-Brami, B. Bussel, and M. A. Maier, "Affected and unaffected quantitative aspects of grip force control in hemiparetic patients after stroke," *Brain Res.*, vol. 1452, pp. 96–107, May 2012, doi: 10.1016/j.brainres.2012.03.007.